Cardiovascular System (CARD) · NPLEX-P1-CARD-08

CV-8 Coronary Circulation

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-08-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the coronary arteries and their territories

A competent response contains

· Both coronary arteries are the first branches of the aorta and arise from the aortic sinuses: the right coronary artery from the right sinus, the left main from the left sinus.

· Each artery runs in a surface groove and supplies the muscle on either side of that groove, so the grooves predict the territories.

· The coronary sulcus circles the heart between atria and ventricles, the anterior and posterior interventricular grooves mark the septum, and the crux is where the coronary sulcus meets the posterior interventricular groove.

· Right coronary artery branches in order: conus branch, sinoatrial nodal artery in about 60% of hearts, right marginal artery at the acute margin, then at the crux the atrioventricular nodal artery in about 80% and the posterior descending artery in right dominant hearts.

· Right coronary artery territory: right atrium, most of the right ventricle, and both nodes in most hearts.

· Left main divides into the left anterior descending and circumflex arteries, and sometimes a ramus intermedius, so its occlusion threatens most of the left ventricle.

· Left anterior descending artery runs in the anterior interventricular groove to the apex, with septal perforators to the anterior two-thirds of the septum carrying the His bundle and both bundle branches, and diagonals to the anterolateral wall; leads V1 to V4.

· Circumflex artery runs in the left coronary sulcus with obtuse marginal branches to the lateral and posterolateral wall and branches to the left atrium, and gives the sinoatrial nodal artery in about 40%; leads I, aVL, V5, V6.

· Posterior descending artery runs in the posterior interventricular groove and supplies the inferior wall, the posterior one-third of the septum, and the posteromedial papillary muscle; leads II, III, aVF.

· Surface relationships: the anterior surface is mostly right ventricle, the inferior surface is mostly left ventricle, and the left surface is the lateral wall.

· The left anterior descending artery is occluded most often in infarction, followed by the right coronary artery and then the circumflex, because it feeds the largest share of left ventricular muscle.

Distinctions that matter

· The sinoatrial nodal percentage (about 60% right coronary) and the atrioventricular nodal percentage (about 80% right coronary) are different figures from different origins, so keep them apart.

· Territory is set by the groove the artery occupies, while dominance is a separate question about which artery reaches the crux and gives the posterior descending artery.

Common misconceptions

· Writing that the left coronary artery supplies the whole left ventricle. The inferior wall and the posterior third of the septum belong to the posterior descending artery, which comes from the right coronary artery in most hearts.

· Assuming the artery most often occluded is the one supplying the most important structures. It is the one supplying the largest mass of left ventricular muscle.

Where it is taught: Section C01, paragraphs on each artery and the table of courses and territories of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C01-BD2

Brain dump 2 · integration and application: Work backward from the tracing to the artery

A competent response contains

· Patient 1, V1 to V4: left anterior descending artery, anterior left ventricle and apex with the anterior two-thirds of the septum, bundle branches and His bundle at risk, so block is infranodal and the escape is wide and slow.

· Patient 2, II, III, aVF: the artery giving the posterior descending artery, usually the right coronary artery, losing the inferior wall and the posterior one-third of the septum, with the atrioventricular node at risk, so block is at the node with a narrow junctional escape; sinus bradycardia is also common because the right coronary artery feeds the sinoatrial node in about 60%.

· Patient 3, I, aVL, V5, V6: circumflex artery, lateral and posterolateral wall and left atrium; conduction is usually spared unless the heart is left dominant, when the circumflex supplies the atrioventricular nodal artery.

· Coronary arteries are functional rather than absolute end arteries, so a stenosis developing over months allows small anastomoses to enlarge into useful collaterals and the muscle beyond can stay alive.

· A sudden thrombus gives no time for collateral growth, so the territory of the blocked artery infarcts.

· The anastomoses are best developed near the apex and within the septum.

· The right ventricle is supplied by the right coronary artery through the conus and right marginal branches, so right ventricular infarction follows a proximal right coronary occlusion and is read in right-sided leads; the left anterior descending artery supplies almost no free right ventricular wall.

Distinctions that matter

· Lead group names the wall, and the wall names the artery, but the conduction consequence depends on which structures sit inside that territory.

· Collaterals protect against a slowly developing stenosis and not against an acute occlusion, which is why the time course matters more than the final degree of narrowing.

Common misconceptions

· Calling the coronary arteries true end arteries. They are functional end arteries, with anastomoses that can enlarge over months.

· Expecting a circumflex occlusion to cause atrioventricular nodal block routinely. It does so only in a left dominant heart.

Where it is taught: Section C01, why the anatomy matters and the final paragraph on collaterals of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-BD1

Brain dump 1 · foundational retrieval: Dominance and the supply of the papillary muscles

A competent response contains

· Coronary dominance describes which artery gives rise to the posterior descending artery, not which artery is larger.

· Right dominant: posterior descending artery from the right coronary artery, in most hearts, about 70% to 85%.

· Left dominant: posterior descending artery from the circumflex, about 5% to 15%.

· Codominant: both the right coronary artery and the circumflex reach the posterior interventricular groove, about 5% to 20%.

· The posterior descending artery supplies the inferior wall of the left ventricle, the posterior one-third of the septum, and the posteromedial papillary muscle.

· The artery that gives the posterior descending artery usually also gives the atrioventricular nodal artery, which is why dominance decides whether an inferior infarct brings nodal block.

· In systole the papillary muscles pull on the chordae tendineae and keep the mitral leaflets from prolapsing into the left atrium.

· The anterolateral papillary muscle lies where left anterior descending diagonal and circumflex obtuse marginal territories overlap, so its supply is dual and it rarely ruptures.

· The posteromedial papillary muscle usually depends on the posterior descending artery alone, so its supply is single and it shares the fate of whichever artery gives the posterior descending artery.

· Papillary tips lie deep in the cavity, at the end of the coronary branches and within the subendocardium, which makes them highly sensitive to ischemia.

Distinctions that matter

· Dominance is about the source of one artery; it is not a statement about how much muscle each coronary artery supplies, since the left coronary artery supplies most of the left ventricle in nearly every heart.

· Single against dual supply, not size or position alone, is what makes one papillary muscle vulnerable and the other safe.

Common misconceptions

· Saying most hearts are left dominant because the left coronary artery supplies more muscle. Most hearts are right dominant because the right coronary artery gives the posterior descending artery.

· Treating both papillary muscles as equally at risk. The posteromedial muscle has one source and the anterolateral has two.

Where it is taught: Section C02, opening paragraph, the dominance table, and blood supply of the papillary muscles of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-BD2

Brain dump 2 · integration and application: Sudden decompensation four days after an inferior infarct

A competent response contains

· The posteromedial papillary muscle has ruptured, supplied by the posterior descending artery, which came from the right coronary artery in this right dominant inferior infarct.

· Dead muscle keeps its tensile strength for about the first day; neutrophils infiltrate and from about day 3 macrophages remove necrotic myocytes before granulation tissue and collagen replace them, so systolic traction tears softened tissue, most often about 2 to 7 days after infarction.

· Sudden mitral incompetence ejects much of each stroke volume into a normal-sized, relatively stiff left atrium, so left atrial pressure rises steeply and pulmonary edema is abrupt.

· Forward stroke volume falls at the same time, which causes hypotension and can progress to cardiogenic shock.

· The murmur can be short and soft because the gradient across the valve collapses before systole ends, so loudness does not track severity here.

· Post-infarct ventricular septal rupture also causes sudden decompensation with a new systolic murmur, and the measurement that separates the two is an oxygen saturation step-up from right atrium to right ventricle, which occurs in septal rupture and not in papillary rupture.

· Ischemic papillary muscle dysfunction without rupture is more common and causes milder regurgitation.

· In a left dominant heart the circumflex gives the posterior descending artery, so the same event follows a circumflex occlusion and the tracing shows an inferolateral infarct, with the atrioventricular node also at risk.

Distinctions that matter

· A chronic stiff left atrium accommodates regurgitation at lower pressure, while an acute lesion in a normal-sized atrium sends pressure straight to the lungs.

· Murmur intensity reflects the pressure gradient and its duration, not the volume of regurgitation.

Common misconceptions

· Expecting rupture during the first hours. Strength is preserved early and falls once enzymatic removal of dead myocytes is under way.

· Using a loud murmur as the sign of severe acute mitral regurgitation. The murmur here is often short and soft.

Where it is taught: Section C02, paragraphs on papillary supply and timing of rupture, the flow diagram, and manifestations explained by mechanism of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-RT01

Retrieval exercise: Rebuild the chain from inferior infarct to mitral regurgitation

A competent response contains

· Step 1: a thrombus occludes the artery giving the posterior descending artery.

· Step 2: the posteromedial papillary muscle loses its only supply, and it is the one at risk because the anterolateral muscle keeps a dual supply from diagonal and obtuse marginal branches.

· Step 3: coagulative necrosis of the papillary muscle.

· Step 4: neutrophils and then macrophages soften it over days 2 to 7, removing necrotic myocytes before collagen can replace them.

· Step 5: systolic traction tears the muscle or one of its heads.

· Step 6: the chordae lose their anchor and the mitral leaflets flail.

· Step 7: acute severe mitral regurgitation, with left atrial pressure rising steeply and forward stroke volume falling.

· In a left dominant heart only the occluded artery changes, from the right coronary artery to the circumflex; the rest of the chain is identical.

Distinctions that matter

· The chain depends on single supply, not on infarct size, which is why the largest infarcts are not the ones that rupture papillary muscle.

· Necrosis and softening are separate stages, and the delay before rupture belongs to the softening stage.

Common misconceptions

· Putting rupture immediately after necrosis. The enzymatic softening step is what sets the day 2 to 7 window.

· Naming the right coronary artery as the cause in every case. The artery that gives the posterior descending artery is the one that matters.

Where it is taught: Section C02, flow diagram from inferior infarct to papillary muscle rupture of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-CP01

Comparison table: Anterolateral against posteromedial papillary muscle

A competent response contains

· Wall: the anterolateral muscle sits on the anterolateral wall, the posteromedial muscle on the inferior wall.

· Supply: the anterolateral muscle is dual, from left anterior descending diagonal and circumflex obtuse marginal branches; the posteromedial muscle is single, usually from the posterior descending artery alone.

· Infarct at risk: an anterolateral infarct threatens the first muscle only if more than one artery is involved, while an inferior infarct threatens the posteromedial muscle directly.

· Rupture frequency: in commonly quoted series the posteromedial muscle ruptures about 6 to 12 times as often as the anterolateral muscle.

· A left anterior descending occlusion leaves the anterolateral muscle perfused by circumflex branches, and the posteromedial muscle lies outside the left anterior descending territory in most hearts, which is why the largest infarcts rarely rupture a papillary muscle.

· In a left dominant heart the posteromedial muscle depends on the circumflex, because the circumflex then gives the posterior descending artery; the anterolateral supply is unchanged.

· Rupture of either muscle removes the anchor for the chordae, the leaflets flail, and acute severe mitral regurgitation follows with abrupt pulmonary edema and falling forward stroke volume.

· Both muscles have tips lying deep in the cavity within the subendocardium, at the end of the coronary branches, so both are sensitive to ischemia even though only one usually ruptures.

Distinctions that matter

· Dual against single supply is the whole explanation for the difference in rupture risk.

· Dominance changes which artery feeds the posteromedial muscle but never makes its supply dual.

Common misconceptions

· Assuming the larger infarct carries the larger rupture risk. Risk follows loss of the only supply.

· Thinking a left dominant heart protects the posteromedial muscle. It simply moves the responsibility to the circumflex.

Where it is taught: Section C02, blood supply of the papillary muscles and the stop and predict box of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C02-CA01

Clinical application: Breathlessness and a new murmur five days after chest pain

A competent response contains

· Recognize: rupture of the posteromedial papillary muscle, supplied by the posterior descending artery from the right coronary artery, causing acute severe mitral regurgitation.

· Explain: that muscle depends on the posterior descending artery alone, so an inferior infarct removes its only supply, and rupture waits until neutrophils and then macrophages have softened the dead muscle from about day 3, giving the usual 2 to 7 day window rather than day one, when tensile strength is still preserved.

· Differentiate: ventricular septal rupture gives the same sudden decompensation with a new systolic murmur but produces a saturation step-up from right atrium to right ventricle, and ischemic papillary dysfunction without rupture is more common and causes milder regurgitation without this degree of pulmonary edema and hypotension.

· Investigate: right atrial 62% and right ventricular 63% show no step-up, which rules out septal rupture; a septal rupture would have shown a clear rise in saturation from right atrium to right ventricle because oxygenated left ventricular blood enters the right ventricle.

· Decide: the subendocardium of the inferior wall became ischemic first, and the ventricular rate of 44/min with dissociated P waves came from block at the atrioventricular node, supplied by the atrioventricular nodal artery from the right coronary artery in about 80% of hearts.

· Justify: the regurgitant volume enters a normal-sized, relatively stiff left atrium, so left atrial pressure rises steeply and pulmonary edema is abrupt, while forward stroke volume falls and causes hypotension; the murmur is short and soft because the gradient across the valve collapses before systole ends.

Distinctions that matter

· The saturation step-up is the single measurement that separates the two post-infarct ruptures.

· Acute mitral regurgitation into a compliant chronically dilated atrium behaves quite differently from the same lesion in a normal atrium.

Common misconceptions

· Grading severity by the loudness of the murmur in acute mitral regurgitation.

· Reading the narrow QRS escape at 44/min as infranodal disease. A narrow escape places the block at the node, which fits the inferior territory.

Where it is taught: Section C02, papillary supply, rupture timing, and manifestations; nodal supply in Section C01 of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C03-BD1

Brain dump 1 · foundational retrieval: Build the cardiac venous drainage table from memory

A competent response contains

· Great cardiac vein: anterior interventricular groove with the left anterior descending artery, then the left coronary sulcus with the circumflex; drains the anterior and lateral left ventricle, anterior septum, and left atrium; becomes the coronary sinus.

· Middle cardiac vein: posterior interventricular groove with the posterior descending artery; drains the inferior wall and posterior septum; enters the right end of the coronary sinus.

· Small cardiac vein: inferior margin with the right marginal artery, then the right coronary sulcus; drains the inferior right ventricle and right atrium; enters the right end of the coronary sinus.

· Anterior cardiac veins: cross the right coronary sulcus from the anterior right ventricle; drain the anterior right ventricle; open directly into the right atrium, bypassing the sinus.

· Smallest cardiac veins (venae cordis minimae): lie within the myocardium; drain the inner myocardium near each chamber; open directly into the chambers, mostly the atria, bypassing the sinus.

· The coronary sinus is a wide, thin-walled channel in the posterior coronary sulcus between the left atrium and left ventricle.

· It opens into the right atrium between the opening of the inferior vena cava and the tricuspid orifice, partly guarded by the valve of the coronary sinus.

· Its ostium forms one border of the triangle that locates the atrioventricular node.

· About 75% of total coronary flow returns through the sinus, mainly from the left ventricle, while most right ventricular venous blood returns through the anterior cardiac veins.

· Each named vein shadows an artery, which is why the venous map can be rebuilt from the arterial map.

Distinctions that matter

· Three veins empty into the coronary sinus and two bypass it, so sinus flow is not the whole of coronary venous return.

· The coronary sinus drains mainly the left ventricle; the right ventricle sends most of its venous blood straight into the right atrium.

Common misconceptions

· Assuming all cardiac venous blood passes through the coronary sinus. About a quarter of it does not.

· Placing the coronary sinus opening at the tricuspid orifice itself. It opens between that orifice and the inferior vena cava.

Where it is taught: Section C03, paragraphs on the coronary sinus and tributaries, and the table of cardiac veins of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C03-BD2

Brain dump 2 · integration and application: Read the venous side as a measurement

A competent response contains

· Coronary sinus blood is about 25% to 30% saturated because the myocardium extracts about 70% of delivered oxygen at rest, far more than the whole body average, so it is among the most deoxygenated blood in the body.

· Arterial blood carries about 20 mL O2/dL and coronary sinus blood about 5 to 6 mL O2/dL.

· A coronary sinus sample reflects left ventricular oxygen extraction and lactate release, because most of what it carries comes from the left ventricle.

· A dilated coronary sinus is explained by a persistent left superior vena cava, which drains into it.

· The sinus begins where the great cardiac vein meets the oblique vein of the left atrium, a remnant of the embryonic left superior vena cava, and the sinus itself develops from the left horn of the sinus venosus.

· The finding is usually harmless because the blood still reaches the right atrium.

· The normal physiological shunt of about 2% of cardiac output comes from smallest cardiac veins opening into the left heart and from bronchial venous blood draining into the pulmonary veins.

· That shunted blood never passes alveoli, so it enters the systemic circulation still deoxygenated and lowers arterial PO2 slightly below alveolar PO2, which is one reason the alveolar to arterial PO2 difference is not zero in a healthy person.

Distinctions that matter

· The low coronary sinus saturation is a normal consequence of high extraction, not a sign of disease.

· A persistent left superior vena cava dilates the sinus without creating a shunt, while the smallest cardiac veins draining into the left heart do create a small true shunt.

Common misconceptions

· Reading a dilated coronary sinus as obstruction to venous drainage. The usual cause is extra inflow from a persistent left superior vena cava.

· Expecting arterial PO2 to equal alveolar PO2 in a healthy person. A small physiological shunt always keeps a gradient.

Where it is taught: Section C03, the coronary sinus paragraph and physiological consequences of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw coronary flow across the cardiac cycle

A competent response contains

· Left coronary flow falls sharply, and can briefly reverse, as systole begins, because intramyocardial pressure in the inner wall rises to about cavity pressure and squeezes the vessels shut.

· Flow surges in early diastole when that compression is released, and about 70% to 80% of left ventricular flow occurs in diastole.

· Resting coronary flow is about 225 mL/min, about 4% to 5% of cardiac output, for an organ that is about 0.5% of body weight.

· Left ventricular coronary perfusion pressure is about aortic diastolic pressure minus left ventricular end-diastolic pressure, because diastolic cavity pressure is the back pressure on the inner vessels.

· With 120/80 mm Hg and an end-diastolic pressure of 8 mm Hg, perfusion pressure is about 80 minus 8, or about 72 mm Hg.

· Hypotension and aortic regurgitation lower the first term; heart failure and a stiff hypertrophied ventricle raise the second.

· The right ventricle is perfused throughout the cycle because its systolic pressure is only about 25 mm Hg, well below aortic pressure; pulmonary hypertension that raises right ventricular pressure toward systemic levels removes that advantage.

· Arterial blood carries about 20 mL O2/dL and coronary sinus blood about 5 to 6 mL O2/dL, so resting extraction is about 70% to 75% for the heart against about 25% for the whole body.

· Four myocyte features forcing high extraction: continuous contraction, almost all ATP from oxidative phosphorylation with mitochondria filling about one-third of cell volume, small creatine phosphate stores with little anaerobic capacity, and about one capillary per myocyte with cytoplasmic myoglobin easing diffusion.

Distinctions that matter

· Aortic systolic pressure drives flow to most organs, while left ventricular coronary flow is driven by aortic diastolic pressure against end-diastolic cavity pressure.

· High extraction is a feature of the resting heart, not a sign of stress, and it is what removes extraction as a reserve.

Common misconceptions

· Drawing peak coronary flow at peak aortic pressure. Peak left coronary flow is in early diastole, when compression is released.

· Using mean arterial pressure in the perfusion pressure estimate. The diastolic value is the relevant one for the left ventricle.

Where it is taught: Section C04, diastolic perfusion of the left ventricle and high oxygen extraction of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-BD2

Brain dump 2 · integration and application: Four patients with ischemia and normal arteries

A competent response contains

· Wall stress is proportional to cavity pressure times radius divided by wall thickness, by the law of Laplace.

· Aortic stenosis: systolic cavity pressure, muscle mass, and end-diastolic pressure all rise, so demand rises and perfusion pressure falls; the failing term is supply against a raised demand.

· Aortic regurgitation at 150/40 mm Hg with an end-diastolic pressure of 25 mm Hg: perfusion pressure is about 40 minus 25, or about 15 mm Hg, against a normal of about 70 mm Hg, and the dilated ventricle has a large radius and high wall stress.

· Heart rate 160/min: diastole shortens far more than systole, so diastolic perfusion time per minute falls while each extra beat raises demand.

· Hemoglobin 6 g/dL: arterial oxygen content falls, and because extraction is already near its limit the fall cuts delivery directly, so flow is the only compensation available.

· In all four patients the subendocardium becomes ischemic first.

· Four reasons the subendocardium loses out: it lies at the end of the penetrating branches that enter from the epicardium, it receives almost no systolic flow, it faces left ventricular end-diastolic pressure in diastole, and it carries the highest wall stress and oxygen demand with the least remaining vasodilator reserve because lower resting tone already spends part of it.

· Hypertrophy lowers wall stress by increasing wall thickness, but capillaries lag behind muscle growth and the stiff wall raises end-diastolic pressure, so the inner wall remains vulnerable.

· Exertion in these patients produces ST depression, because ischemia is confined to the inner layer rather than being transmural.

Distinctions that matter

· Ischemia can arise with normal coronary arteries whenever perfusion pressure, perfusion time, or oxygen content falls, or demand rises.

· ST depression points to inner wall ischemia while ST elevation points to transmural injury, so the pattern follows the layer involved.

Common misconceptions

· Assuming chest pain on exertion always means a coronary stenosis. Valve disease, tachycardia, hypertrophy, and anemia all produce it.

· Treating hypertrophy as protective because it reduces wall stress. Capillary density and end-diastolic pressure move the wrong way.

Where it is taught: Section C04, subendocardial vulnerability, the Laplace paragraph, and the paragraph on ischemia with normal arteries of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-RT01

Retrieval exercise: Rebuild the tachycardia sequence with the arithmetic

A competent response contains

· Step 1: heart rate rises, from exercise, fever, anemia, or arrhythmia.

· Step 2: diastole shortens far more than systole.

· Step 3: diastolic perfusion time per minute falls.

· Step 4: each extra beat raises oxygen demand, because every contraction costs oxygen and more contractions occur each minute.

· Step 5: the subendocardium exhausts its vasodilator reserve.

· Step 6: inner wall ischemia appears as angina with ST depression.

· At 70/min the cycle lasts about 0.86 s, systole about 0.30 s and diastole about 0.56 s, giving about 39 s of diastole per minute.

· At 150/min the cycle lasts 0.40 s, systole shortens only to about 0.22 s and diastole is about 0.18 s, giving about 27 s of diastole per minute, a fall of about 30%.

· Tachycardia lowers supply and raises demand at the same time, which is why it is poorly tolerated once a stenosis limits the rise in flow.

Distinctions that matter

· Systole shortens only slightly as rate rises, so almost the whole reduction in cycle length comes out of diastole, which is the perfusion window for the left ventricle.

· Perfusion time per minute, not the length of a single diastole, is the quantity that matters for total flow.

Common misconceptions

· Assuming systole and diastole shorten proportionally. Diastole takes almost all the loss.

· Thinking a faster rate helps coronary flow by increasing the number of diastolic intervals. Total diastolic seconds per minute fall.

Where it is taught: Section C04, flow diagram on tachycardia and the worked example on diastolic time per minute of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-PR01

Prediction exercise: Predict the effect of a fast atrial rhythm in aortic stenosis

A competent response contains

· Diastole in each cycle shortens, and total diastolic seconds per minute fall, because systole shortens only slightly as rate rises.

· Left ventricular end-diastolic pressure rises, because a stiff hypertrophied ventricle filling in less time needs a higher pressure for the same volume.

· Coronary perfusion pressure falls, since it is about aortic diastolic pressure minus end-diastolic pressure and the second term has risen.

· Oxygen demand rises, because wall stress is proportional to cavity pressure times radius divided by wall thickness and aortic stenosis keeps systolic cavity pressure high, while the extra beats each cost oxygen.

· Subendocardial vasodilator reserve falls toward zero, because lower resting tone in that layer has already spent part of it.

· The subendocardium becomes ischemic first and the electrocardiogram shows ST depression.

Distinctions that matter

· Rate affects supply and demand in the same direction of harm, which is why a fast rhythm is dangerous in aortic stenosis.

· Hypertrophy reduces wall stress but raises end-diastolic pressure, so it does not protect the inner wall.

Common misconceptions

· Expecting metabolic vasodilation to make up the difference. Reserve in that layer is partly spent at rest and runs out first.

· Looking for ST elevation. Inner wall ischemia without transmural injury gives ST depression.

Where it is taught: Section C04, subendocardial vulnerability, the tachycardia flow diagram, and the paragraph on aortic valve disease of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C05-BD1

Brain dump 1 · foundational retrieval: Rebuild metabolic control of coronary flow

A competent response contains

· By the Fick principle, myocardial oxygen consumption equals coronary flow times the arteriovenous oxygen content difference.

· Skeletal muscle can widen that difference in exercise, but the myocardium already extracts about 70% to 75% at rest and can reach only about 80% to 90%, so flow is the only term left to change.

· Adenosine pathway: oxygen demand exceeds supply; ATP breakdown outpaces regeneration and AMP accumulates; 5 prime nucleotidase converts AMP to adenosine; adenosine binds arteriolar A2A receptors coupled to Gs; cAMP rises and activates protein kinase A; K+ channels open, including ATP-sensitive channels; hyperpolarization closes L-type Ca2+ channels, cytosolic Ca2+ falls, myosin light chain kinase activity drops, the arteriole relaxes, resistance falls, and flow rises until supply matches demand.

· Adenosine is cleared within seconds by cellular uptake and adenosine deaminase, which makes this a fast negative feedback loop.

· Other mediators acting in parallel: low PO2 and ATP opening ATP-sensitive K+ channels directly, CO2, H+, lactate, K+ released by repeated depolarization, and endothelial nitric oxide released by shear stress.

· Blocking adenosine receptors blunts but does not abolish the rise in flow, which is the evidence that other mediators contribute.

· Sympathetic alpha-1 constriction is mild, while beta-1 stimulation raises cardiac work and metabolite release, so sympathetic activation raises coronary flow overall.

· Autoregulation holds flow nearly constant over a perfusion pressure range of about 60 to 140 mm Hg.

· Coronary flow reserve is maximal flow during full vasodilation divided by resting flow, normally about 3 to 5.

Distinctions that matter

· Local metabolic control overrides neural control in this bed, which is why beta-1 effects on work raise flow despite alpha-1 constriction.

· Adenosine is the clearest single mediator but not the only one, so receptor blockade blunts rather than abolishes the response.

Common misconceptions

· Expecting the heart to meet higher demand by extracting more oxygen. Extraction is nearly maximal at rest.

· Treating sympathetic activation as a coronary vasoconstrictor overall. The rise in work and metabolite release dominates the mild alpha-1 effect.

Where it is taught: Section C05, opening paragraphs, the adenosine flow diagram, and the paragraph on parallel mediators of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C05-BD2

Brain dump 2 · integration and application: Work the numbers on a stenosis, then explain steal

A competent response contains

· Arteriovenous difference is 20 minus 6, which is 14 mL O2/dL.

· Resting consumption is 2.25 dL/min times 14 mL/dL, about 32 mL O2/min, and extraction is 14 divided by 20, which is 70%.

· Doubling demand gives about 63 mL O2/min; raising extraction to about 85%, or 17 mL/dL, at unchanged flow yields only about 38 mL/min.

· Meeting the new demand at the resting extraction needs 63 divided by 14, about 4.5 dL/min, which is double the resting flow, so any lesion that caps flow causes ischemia under stress.

· With a 75% diameter stenosis, resting flow in that territory is normal because the downstream arterioles dilate to offset the stenosis, so arteriolar tone is already low and part of the flow reserve is spent at rest, while maximal flow is reduced.

· Maximal flow starts to fall at about 50% diameter narrowing, and resting flow is preserved until about 80% or more.

· The patient is comfortable sitting because resting flow is adequate, and has pain climbing stairs because the arterioles are already dilated, maximal flow falls short of demand, and the subendocardium becomes ischemic first, which is the mechanism of stable angina.

· Given a vasodilator, normal beds dilate while the bed beyond the stenosis is already maximally dilated, so flow shifts toward normal myocardium and can fall in the ischemic bed, which is coronary steal.

· The notes use adenosine and dipyridamole, which blocks cellular adenosine uptake and so raises adenosine levels, and this uneven redistribution is what makes a perfusion defect visible during pharmacologic stress imaging.

Distinctions that matter

· Resting flow and maximal flow fail at different degrees of narrowing, which is why a patient can be symptom free at rest with a severe lesion.

· Steal is a redistribution of flow rather than a fall in total coronary flow.

Common misconceptions

· Reading normal resting flow as evidence of a mild stenosis. Normal resting flow is maintained by spending reserve.

· Expecting a vasodilator to help the ischemic bed. The stenosed bed cannot dilate further, so flow moves away from it.

Where it is taught: Section C05, the Fick worked example and the paragraphs on autoregulation, flow reserve, and steal of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild why ALCAPA fails only after birth

A competent response contains

· In ALCAPA the left main coronary artery arises from the pulmonary trunk while the right coronary artery arises normally from the aorta.

· Before birth, high pulmonary vascular resistance and the ductus arteriosus keep pulmonary artery pressure about equal to aortic pressure, so the anomalous artery fills antegrade at systemic pressure.

· The fetus tolerates it because pulmonary artery blood is only modestly less saturated than ascending aortic blood and the fetal myocardium is adapted to low oxygen tension, so the heart develops normally.

· After birth, pulmonary vascular resistance falls, the ductus closes, and pulmonary artery pressure falls over days to weeks toward about 25/10 mm Hg.

· The left coronary artery then carries low-pressure blood about 70% to 75% saturated, which cannot supply a left ventricle perfused mainly in diastole against end-diastolic pressure and extracting about 70% of its oxygen.

· Collaterals from the right coronary artery, still at aortic pressure, fill the left coronary branches, but because the pulmonary trunk is the lowest-pressure outlet, blood flows backward up the left coronary artery into it, so the artery becomes a drain.

· That is a form of coronary steal, and it leaves the left ventricle ischemic despite growing collateral flow, with anterolateral ischemia and infarction often at 2 to 3 months.

· Self-reinforcing cycle: low-pressure flow and steal leave the ventricle ischemic; the ischemic ventricle contracts poorly and dilates; the larger radius raises wall stress and demand; papillary ischemia and annular dilation cause mitral regurgitation; end-diastolic pressure rises and perfusion pressure falls further; ischemia deepens and drives further dilation.

· Each step in that cycle either raises demand (larger radius and wall stress, regurgitant volume load) or lowers supply (higher end-diastolic pressure, lower perfusion pressure), which is why the infant ventricle can deteriorate quickly.

Distinctions that matter

· The anomalous origin is present from the start, and what changes after birth is pulmonary artery pressure, so the lesion fails because of the normal birth transition.

· Collateral development is not protective here, because the collateral flow drains into the low-pressure pulmonary trunk.

Common misconceptions

· Expecting the infant to be ischemic from birth. Pulmonary artery pressure is still high in the first days.

· Assuming growing right coronary collaterals will rescue the left ventricle. They feed the steal.

Where it is taught: Section C06, paragraphs on fetal and postnatal physiology, the flow diagram, and the self-reinforcing cycle of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C06-BD2

Brain dump 2 · integration and application: The infant who looks well at birth and sick at ten weeks

A competent response contains

· Symptoms begin around 2 to 3 months because that is when pulmonary artery pressure has reached its low postnatal level, and not in the first days, when it is still high.

· Feeding is an infant's main exertion, so ischemia appears with feeds or crying as pallor, sweating, irritability, and breathlessness, described as infant angina.

· Examination and imaging show heart failure: tachypnea, poor weight gain, and cardiomegaly on the chest film, with a dilating poorly contracting ventricle.

· The electrocardiogram shows an anterolateral infarct pattern with abnormal Q waves and ST and T changes in leads I, aVL, V5, and V6, which is almost unheard of in a normal infant.

· Mitral regurgitation arises from anterolateral papillary muscle ischemia and from annular dilation.

· What separates this from dilated cardiomyopathy and myocarditis is a regional infarct pattern in the left coronary territory rather than global dysfunction, and onset after the first weeks rather than from birth.

· A minority develop extensive right coronary collaterals and survive into childhood or adulthood with a dilated right coronary artery, a continuous murmur, mitral regurgitation, exercise-induced ischemia, or sudden arrhythmic death.

· In those patients pulmonary artery saturation can be slightly higher than right ventricular saturation, because the coronary system creates a small left to right shunt into the pulmonary trunk.

· A large ventricular septal defect keeps pulmonary artery pressure high, which can delay or mask the ischemia.

· The variable that decides perfusion is pulmonary artery pressure relative to the pressure needed to perfuse the left ventricle, not the anomalous origin itself.

Distinctions that matter

· Regional against global dysfunction is what points to a coronary cause rather than a myopathic or inflammatory one.

· Most infants deteriorate in the first year, while a minority with rich collaterals present much later, so the same anatomy gives two very different courses.

Common misconceptions

· Reading a normal first few weeks as evidence against the diagnosis. The timing is the expected pattern.

· Assuming pulmonary artery saturation can never exceed right ventricular saturation. The coronary shunt into the trunk makes it possible here.

Where it is taught: Section C06, manifestations explained by mechanism and variability and distinctions of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C01-Q01

Board-style question: Locating the occlusion from the complication

Correct answer: C. Right coronary artery supplying the atrioventricular node

Why it is right: The right coronary artery runs to the crux, where it gives the atrioventricular nodal artery in about 80% of hearts and the posterior descending artery in right dominant hearts, so its occlusion produces an inferior infarct in II, III, and aVF together with nodal block and a narrow junctional escape at 40 to 60 bpm.

A is wrong: A left anterior descending occlusion shows in V1 to V4 and damages the bundle branches, giving a wide escape at about 20 to 40 bpm rather than a 0.08 s QRS.

B is wrong: A circumflex occlusion shows in I, aVL, V5, and V6 and spares the atrioventricular node unless the heart is left dominant, which the inferior lead pattern here does not support.

D is wrong: A left main occlusion threatens most of the left ventricle and gives widespread anterior and lateral changes with profound hypotension, not isolated inferior ST elevation.

A competent response contains

· ST elevation in II, III, and aVF marks the inferior wall, the territory of the posterior descending artery, which comes from the right coronary artery in most hearts.

· P waves at 82/min with dissociated QRS complexes at 46/min of 0.08 s duration are complete block with a narrow junctional escape, which places the block at the atrioventricular node, supplied by the atrioventricular nodal artery from the right coronary artery in about 80% of hearts.

Distinctions that matter

· A narrow escape complex places the pacemaker above the bundle branches and so places the block at the node.

Common misconceptions

· Reading every complete block after infarction as infranodal. The lead group and the narrow escape point to the node.

Where it is taught: Section C01 of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C04-Q02

Board-style question: Ischemia with unobstructed arteries

Correct answer: A. A coronary perfusion pressure of about 14 mm Hg with high wall stress in a dilated ventricle, which exhausts subendocardial reserve

Why it is right: Her aortic diastolic pressure of 40 mm Hg against an end-diastolic pressure of 26 mm Hg leaves a perfusion pressure of about 14 mm Hg, and the dilated ventricle has high wall stress and oxygen demand, so the subendocardium, which already spends part of its vasodilator reserve at rest and receives almost no systolic flow, becomes ischemic on exertion.

B is wrong: Myocardial extraction is about 70% to 75% at rest rather than 25%, so a failure to rise above 25% does not describe this heart; the whole-body figure is the one near 25%.

C is wrong: The subepicardium is the layer that keeps perfusion best, and it is the subendocardium that receives almost no systolic flow, so this reverses the gradient within the wall.

D is wrong: Diversion of flow into the pulmonary trunk belongs to an anomalous left coronary artery arising from the pulmonary artery, a congenital lesion of infancy, not to aortic regurgitation.

A competent response contains

· Coronary perfusion pressure is about aortic diastolic pressure minus left ventricular end-diastolic pressure, so 40 minus 26 is about 14 mm Hg against a normal of about 70 mm Hg.

· Aortic regurgitation lowers aortic diastolic pressure and dilates the ventricle, and by the law of Laplace the larger radius raises wall stress and oxygen demand, so the subendocardium exhausts its reserve first.

Distinctions that matter

· Ischemia follows the balance of perfusion pressure, perfusion time, oxygen content, and demand, and a normal angiogram does not exclude it.

Common misconceptions

· Requiring a stenosis before accepting ischemia. Valve disease alone can lower supply and raise demand enough to produce it.

Where it is taught: Section C04 of the CV-8 Coronary Circulation notes.

NPLEX-P1-CARD-08-01-C06-Q03

Board-style question: Why the infant deteriorates at two months

Correct answer: D. A fall in pulmonary vascular resistance and pulmonary artery pressure, which left the anomalous artery with low-pressure desaturated blood and reversed collateral flow into the pulmonary trunk

Why it is right: The postnatal fall in pulmonary vascular resistance and closure of the ductus drop pulmonary artery pressure over days to weeks, leaving the anomalous artery with low-pressure desaturated blood, and because the pulmonary trunk is then the lowest-pressure outlet, right coronary collateral flow runs backward up the left coronary artery into it, so the left ventricle stays ischemic and infarcts in the anterolateral territory.

A is wrong: Atherosclerosis does not develop over weeks in an infant, and the lesion here is a congenital origin rather than a narrowing.

B is wrong: The foramen ovale carries blood from right atrium to left atrium and plays no part in perfusing the left ventricular muscle, which depends on the coronary arteries.

C is wrong: Myocardial extraction is already near maximal and does not rise to create the problem; the failure is in the pressure and oxygen content delivered to the anomalous artery.

A competent response contains

· Before birth, pulmonary artery pressure about equals aortic pressure, so the anomalous left coronary artery fills antegrade at systemic pressure with only modestly desaturated blood.

· After birth, pulmonary vascular resistance falls and the ductus closes, so pulmonary artery pressure falls toward about 25/10 mm Hg over days to weeks, the left coronary artery receives low-pressure blood about 70% to 75% saturated, and right coronary collateral flow drains backward into the pulmonary trunk as coronary steal, giving an anterolateral infarct pattern in I, aVL, V5, and V6 at about 2 to 3 months.

Distinctions that matter

· The anatomy does not change after birth; pulmonary artery pressure does, and that is what decides whether the left ventricle is perfused.

Common misconceptions

· Treating collateral growth as protective here. Collateral flow drains into the low-pressure pulmonary trunk instead of perfusing muscle.

Where it is taught: Section C06 of the CV-8 Coronary Circulation notes.